

Joseph Milliken

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EDUCATION

Bachelor of Science in Computer Science and Engineering

08/2022 – 05/2026 | Denver, CO

University Of Colorado Denver

- Relevant Coursework: Algorithms, Data Structures, Software Engineering

GPA: 3.94/4

PROFESSIONAL EXPERIENCE

Teaching Assistant | CSCI3412 Algorithms & CSCI3415 Prin. of Prgm. Lang.

08/2024 – Present | Denver, CO

University of Colorado Denver

- Guide instruction in complex algorithms, with an emphasis on computational complexity, dynamic programming, and greedy algorithms, as well as fundamental programming language concepts.
- Conducted office hours to offer personalized assistance, enhancing students' grasp of algorithm and programming language principles.

PROJECTS

Clara | Capstone Project

Python, React, MongoDB

- Designing and implementing an AI pipeline for whole-slide image (WSI) analysis using the CLAM framework to classify biopsies and generate attention based heatmaps
- Building a Python modeling workflow using PyTorch and integrating inferencing results into a REST API that interfaces with a React-based frontend, enabling Pathologists to review predictions and interpretability maps
- Collaborating in a 4-person team over a two-semester timeline to develop a working product that supports clinical decision-making

bullyometer

FastAPI, BERT

- Developed a FastAPI backend to serve low-latency inference requests from the trained NLP model
- Built a React frontend to display chat activity and real-time detection results, integrating with Supabase subscription channels
- Implemented training, evaluation, and inference pipelines using Hugging Face Transformers, TensorFlow/PyTorch, and Scikit-learn

taskctl

Golang, JSON

- Developed a lightweight, cross-platform command-line task manager
- Implemented core task operations: add, update, delete, mark complete/incomplete, and clear all tasks
- Enhanced usability with features like priority and due dates, sorting by status/priority/date
- Designed a polished interactive UI with promptui, supporting intuitive keyboard navigation

OrbitTrckr

React, Three.js, NASA Open APIs

- Developed a dynamic web application to visualize Near-Earth Objects (NEOs) using three.js, displaying real-time data
- Implemented using React with Tailwind CSS and PostCSS for a responsive, modern, and maintainable user interface
- Integrated NASA NeoW API to fetch and process NEO data within specified timeframes

SKILLS

Languages

Python, C/C++, Golang, C#, SQL,
Javascript, CSS, HTML

Developer Tools/Operating Systems

Git/Github, VS Code, Visual Studio,
Arduino IDE, Unix/Linux, macOS,
Windows

Frameworks/Libraries

React, numpy, pandas

AWARDS

Certifications

NVIDIA Applications of AI for Anomaly Detection
CU Denver Cybersecurity and Secure Computing (CSSC)

Dean's List

Fall 22 | Spring 23 | Fall 23 | Spring 24 | Fall 24 | Spring 25 | Fall 25